

CHEBOKSARY, Russian Federation

WMO: 275810

Lat: 56.083N

Lon: 47.333E

Elev: 536

StdP: 14.41

Time Zone: 3.00 (E03)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.3	-9.3	-19.8	1.9	-13.5	-14.7	2.5	-9.1	24.0	21.7	22.3	21.5	6.1	80	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	88.1	66.0	84.2	65.6	80.7	64.7	69.5	79.7	67.8	78.5	66.6	77.7	7.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	98.6	73.8	64.5	92.6	72.2	62.7	86.9	70.7	33.9	79.3	32.6	78.5	31.5	77.3	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.0	19.8	17.9	DB	-17.0	90.4	4.8	4.8	-20.5	93.8	-23.3	96.6	-26.0	99.3	-29.5	102.8	
			WB	-18.2	72.1	5.2	2.5	-22.0	73.9	-25.0	75.3	-27.9	76.8	-31.7	78.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.0	10.9	14.9	24.5	41.9	58.0	63.8	68.3	65.1	53.7	40.4	29.5	19.8
	DBStd	21.80	11.99	11.50	8.40	8.22	8.48	7.65	6.65	7.97	7.44	8.17	8.23	11.36
	HDD50	5191	1211	983	789	270	25	1	0	1	45	313	615	938
	HDD65	9097	1676	1403	1254	694	248	114	38	96	344	762	1065	1403
	CDD50	1922	0	0	0	26	272	414	568	470	157	15	0	0
	CDD65	354	0	0	0	0	30	77	141	100	6	0	0	0
	CDH74	3099	0	0	0	1	279	685	1236	859	39	0	0	0
	CDH80	959	0	0	0	0	44	202	409	302	2	0	0	0
Wind	WSAvg	9.2	9.8	10.2	10.5	9.7	8.9	8.3	7.1	7.2	7.9	9.9	10.2	10.3
Precipitation	PrecAvg	22.3	1.5	1.1	1.3	1.2	1.5	2.6	2.8	2.4	2.2	2.2	1.8	1.5
	PrecMax	26.1	3.0	2.2	2.8	2.8	2.8	6.3	5.7	4.6	4.5	4.3	3.3	2.4
	PrecMin	15.8	0.6	0.4	0.2	0.2	0.4	0.7	0.4	0.1	0.6	0.4	0.3	0.4
	PrecStd	2.6	0.6	0.5	0.6	0.7	0.6	1.5	1.3	1.1	1.1	1.0	0.7	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	35.7	44.5	71.8	84.3	89.8	95.3	95.2	78.7	64.6	50.4	41.3
		MCWB	33.6	33.7	38.2	55.1	62.4	63.5	65.7	66.6	61.5	54.5	48.0	40.9
	2%	DB	33.4	33.9	39.5	66.2	80.6	85.6	89.2	87.7	73.7	58.8	46.8	37.5
		MCWB	33.0	33.0	34.9	51.1	61.3	67.3	65.9	65.5	59.5	52.0	45.2	36.8
	5%	DB	31.7	32.3	37.4	60.7	77.1	82.0	84.1	82.3	69.6	55.1	44.3	35.2
		MCWB	31.2	31.6	34.1	48.5	59.6	65.3	65.7	65.3	58.3	49.6	42.7	34.5
	10%	DB	28.2	30.3	35.5	55.7	73.2	77.3	80.6	78.6	65.8	51.8	39.6	33.5
		MCWB	27.6	29.4	33.3	45.3	58.4	63.1	65.3	64.2	56.4	47.8	38.5	32.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.3	34.1	38.7	55.5	65.9	71.9	71.2	71.1	64.0	55.8	48.8	41.1
		MCDB	33.5	34.4	43.0	68.1	78.6	82.9	81.2	79.2	73.5	61.2	49.9	41.2
	2%	WB	32.3	32.8	35.8	53.1	63.7	69.6	69.2	68.7	61.5	53.0	45.2	36.5
		MCDB	32.6	33.3	37.8	63.2	76.7	79.9	79.4	79.5	70.2	57.8	46.7	36.8
	5%	WB	30.5	31.6	34.5	49.8	61.4	67.2	67.7	67.2	59.5	50.7	42.1	34.0
		MCDB	30.8	32.3	36.3	58.9	72.6	77.1	78.0	78.9	67.1	54.0	43.3	34.3
	10%	WB	27.0	29.3	33.2	46.5	59.3	65.0	66.6	65.6	57.6	48.4	38.6	32.4
		MCDB	27.6	30.1	35.0	53.7	70.7	75.0	77.7	76.3	63.6	51.2	39.4	32.7
Mean Daily Temperature Range	5% DB	MDBR	9.8	10.2	11.1	14.6	18.7	17.2	17.4	16.8	13.7	9.8	6.9	8.0
		MCDBR	7.8	7.9	11.4	22.4	22.7	21.4	22.2	21.7	20.0	14.7	9.1	6.2
		MCWBR	7.9	7.8	8.5	12.1	10.6	9.6	8.8	8.4	9.5	9.5	8.7	6.3
	5% WB	MCDBR	8.2	7.4	10.0	21.2	20.6	19.5	17.9	18.7	17.3	12.8	8.7	6.0
		MCWBR	8.3	7.5	8.2	11.8	10.8	9.8	8.9	8.5	9.2	9.7	8.7	6.2
Clear-Sky Solar Irradiance	taub		0.285	0.316	0.355	0.402	0.380	0.375	0.399	0.423	0.369	0.347	0.309	0.281
	taud		2.081	2.067	2.034	2.143	2.346	2.399	2.338	2.245	2.383	2.339	2.215	2.062
	Ebn at Noon		186	228	249	257	273	276	265	248	248	220	179	153
	Edh at Noon		20	30	40	42	36	35	36	37	29	23	17	16
All-Sky Solar Radiation	RadAvg		193	475	951	1297	1731	1842	1818	1417	863	397	155	106
	RadStd		29	82	102	131	117	151	164	137	108	62	28	26

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	+0.95	N/A	N/A	N/A	N/A	N/A	-256	-346	N/A	N/A

Nomenclature: See separate page